

# Local Decoder.

David Ponarovsky

December 25, 2025

# Introduction

Today:

- ▶ Proof.

1. Local Decoders.

# Our Code

## Code

In this session, we consider LDPC codes, where each bit adjoins to  $\Delta_1$  checks, and each check adjoins to  $\Delta_2$  bits. We denote by  $\Delta = \Delta_1 \Delta_2$ .

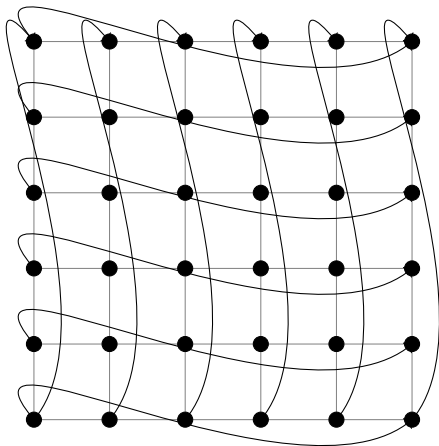
# Local Decoder

## definition

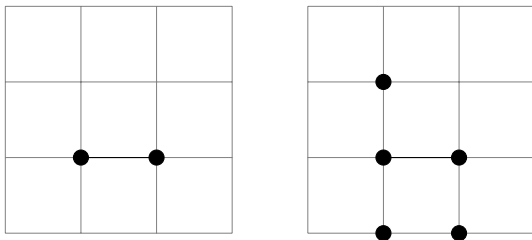
We say that  $D$  is a local decoder if when applied against  $2p(1-p)$ -depolarized channel:

- ▶ There exists function  $M : (0, 1) \rightarrow (0, 1)$  such that the probability of each bit being flipped is  $M(2p(1-p)) < p^{\frac{100}{99}}$ .
- ▶ There is a constant  $l$  such that for any bit, there are only  $(\Delta)^l$  bits on which the event of it being flipped depends.

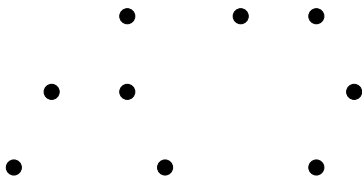
For example, if we take  $k$  large enough and consider the  $(n, k)$ -Toric, then both the swift-rule and the flip upon majority are local decoders.



**Figure:** On the left is the Toric Graph. On the right are cross and face checks.



**Figure:** On the left is the Toric Graph. On the right are cross and face checks.



**Figure:** On the left is the Toric Graph. On the right are cross and face checks.